

Failure Rate Calculation for Contactors

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Name = Contactors

Type = 1-pole DC contactors

Reference = None

Duration = 420

Failure rate $\lambda = \lambda_{\text{ref}} * \text{Pi_S} * \text{Pi_U} * \text{Pi_I} * \text{Pi_T} * \text{Pi_E}$

$\lambda_{\text{ref}} = 200.0$

$\text{Pi_S} = 1$

$\text{Pi_U} = 1.2$

Operating voltage U in V = 440

$\text{Pi_I} = 0.15$

Current ratio (I/Ie) = 0.33

$\text{Pi_T} = 1$

$\text{Pi_E} = 1$

Failure Rate $\lambda = 36.0$

Reliability Calculation for Contactors =

$R(t) = e^{\{-\lambda t\}}$

Reliability = 0.9999848801143066